Zanele Madondo

PRINCIPAL-LEVEL ENGINEER

Software Architecture • Distributed Platforms • Engineering Enablement

Cape Town, South Africa

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Leadership Statement

Software engineer and technical leader with 12+ years of experience delivering resilient systems for payments, logistics, and cloud products. Brings deep implementation capability alongside architecture governance, operational improvement, and sustained mentorship of senior engineers.

Career Experience

Principal Software Engineer 2021 – Present

Atlantic Payment Network, Cape Town

- Defined technical strategy for a 25-engineer transaction platform responsible for more than ZAR 74 billion in annual payments across Southern Africa.
- Guided an incremental shift from a monolithic authorisation system to independently deployable domain services, cutting p99 response time by 41% and moving releases from monthly to several times per week.
- Established regional failover patterns, replicated-data safeguards, and recovery exercises that supported 99.99% availability for business-critical payment APIs.
- Put in place engineering RFCs, architecture reviews, and reliability measures now used by six product domains.
- Supported the growth of five senior engineers; two subsequently took on staff-level technical leadership scope.

Staff Software Engineer 2017 – 2021

Ubuntu Freight Technologies, Durban

- Designed a streaming shipment-visibility platform processing approximately 18 million location events daily with Go, Kafka, Kubernetes, and Cassandra.
- Delivered capacity forecasting and autoscaling controls that lowered yearly infrastructure costs by 28% without sacrificing seasonal performance.
- Collaborated across product, operations, and analytics to introduce predictive exception processes, reducing manual case work by 30%.
- Advanced incident learning and telemetry practices, improving recovery time for tier-one services by 38%.

Senior Software Engineer 2014 – 2017

Karoo Cloud Solutions, Stellenbosch

- Developed multi-tenant provisioning services and engineering tools adopted by over 40 internal product teams.
- Scaled automated testing, service observability, and threat-modelling standards across a growing cloud platform organisation.

Signature Contributions

Platform renewal	Orchestrated a multi-year migration plan that decommissioned 14 legacy services with no major customer-impacting outage.
Operational maturity	Developed incident-command coaching and structured learning reviews, reducing recurring severe incidents by 52%.
Engineering flow	Sponsored a standard service and CI/CD pathway that reduced median time to production readiness from 10 days to 2 days.

Technical Scope

Languages: Go, Java, Python, TypeScript, SQL — **Cloud:** AWS, Kubernetes, Terraform, Docker
Data & communication: Kafka, gRPC, PostgreSQL, DynamoDB, Redis, OpenTelemetry
Leadership practice: Distributed-systems design, SRE, API governance, security reviews, technical planning, mentoring

Education & Industry Activity

MSc Software Engineering, University of Cape Town 2013 – 2014
BSc Computer Science, Stellenbosch University 2009 – 2013

Engineering Excellence Award (2024) • Co-inventor, adaptive workload-routing patent (2022)
Speaker: “Reliable by Design: Payment Platforms Under Load,” Cape Town Engineering Forum (2025)